IEC 61784-5-20

Edition 1.0 2018-08

INTERNATIONAL STANDARD

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Industrial communication networks – Profiles –
Part 5-20: Installation of fieldbuses – Installation profiles for CPF 20

IEC 61784-5-20:2018-08(en)

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Industrial communication networks – Profiles – Part 5-20: Installation of fieldbuses – Installation profiles for CPF 20

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CONTENTS

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INDUSTRIAL COMMUNICATION NETWORKS – PROFILES –

Part 5-20: Installation of fieldbuses – Installation profiles for CPF 20

FOREWORD

International Standard IEC 61784-5-20 has been prepared by subcommittee 65C: Industrial networks, of IEC technical committee 65: Industrial-process measurement, control and automation.

This standard is to be used in conjunction with IEC 61918:2018.

The text of this International Standard is based on the following documents:

FDIS	Report on voting
65C/924/FDIS	65C/925/RVD

Full information on the voting for the approval of this International Standard can be found in the report on voting indicated in the above table.

This document has been drafted in accordance with the ISO/IEC Directives, Part 2.

A list of all parts of the IEC 61784-5 series, published under the general title *Industrial communication networks – Profiles – Installation of fieldbuses*, can be found on the IEC website.

The committee has decided that the contents of this document will remain unchanged until the stability date indicated on the IEC web site under "http://webstore.iec.ch" in the data related to the specific document. At this date, the document will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

IMPORTANT – The 'colour inside' logo on the cover page of this publication indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

INTRODUCTION

This International Standard is one of a series produced to facilitate the use of communication networks in industrial control systems.

IEC 61918:2018 provides the common requirements for the installation of communication networks in industrial control systems. This installation profile standard provides the installation profiles of the communication profiles (CP) of a specific communication profile family (CPF) by stating which requirements of IEC 61918 fully apply and, where necessary, by supplementing, modifying, or replacing the other requirements (see Figure 1).

For general background on fieldbuses, their profiles, and relationship between the installation profiles specified in this document, see IEC 61158-1. Each CP installation profile is specified in a separate annex of this document. Each annex is structured exactly as the reference standard IEC 61918 for the benefit of the persons representing the roles in the fieldbus installation process as defined in IEC 61918 (planner, installer, verification personnel, validation personnel, maintenance personnel, administration personnel). By reading the installation profile in conjunction with IEC 61918, these persons immediately know which requirements are common for the installation of all CPs and which are modified or replaced. The conventions used to draft this document are defined in Clause 5. The provision of the installation profiles in one standard for each CPF (for example IEC 61784-5-20 for CPF 20) allows readers to work with standards of a convenient size.

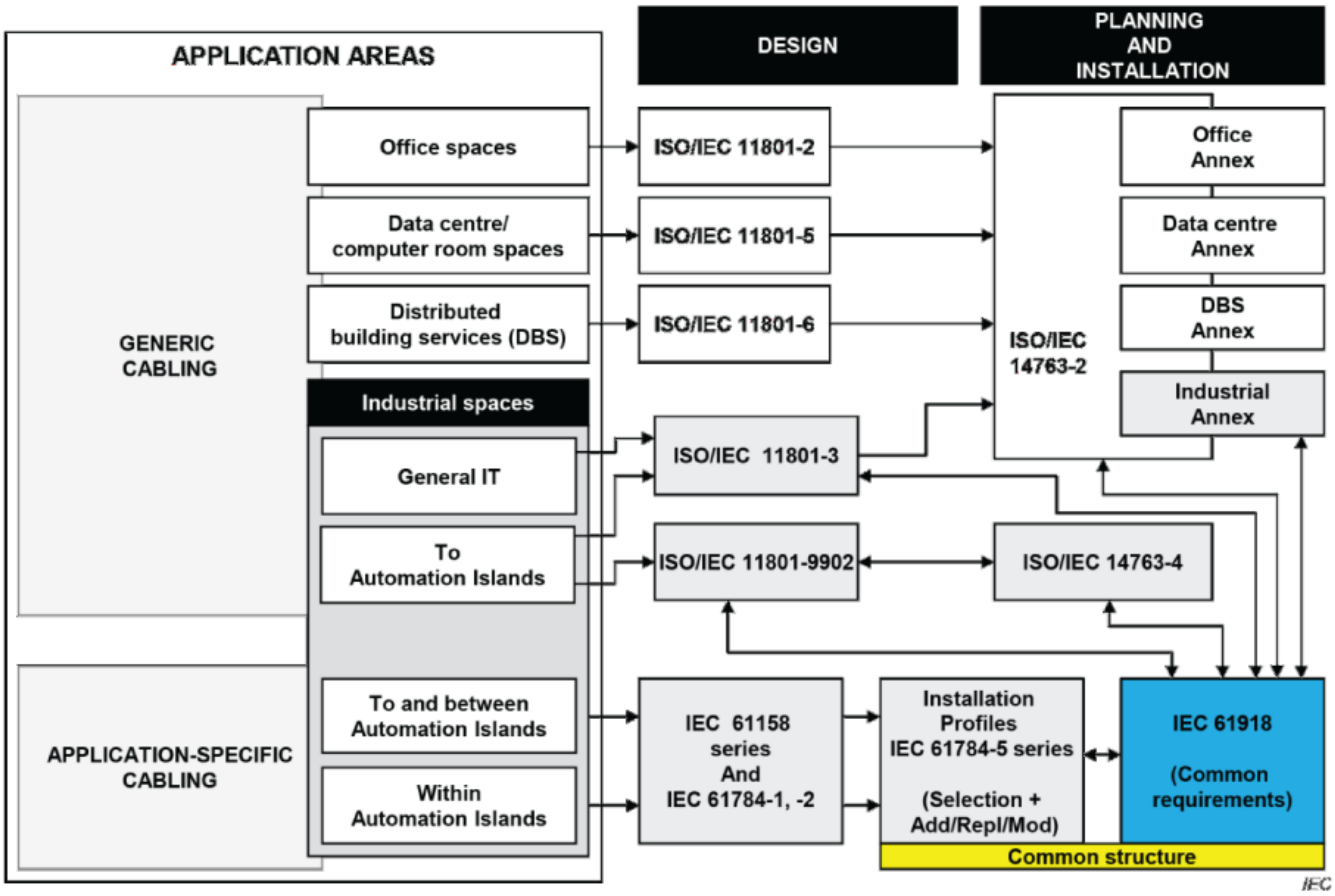

Figure 1 – Standards relationships

INDUSTRIAL COMMUNICATION NETWORKS – PROFILES –

Part 5-20: Installation of fieldbuses – Installation profiles for CPF 20

1 Scope

This part of IEC 61784 specifies the installation profiles for CPF 20 (ADS-net[1])·.

The installation profiles are specified in the annexes. These annexes are read in conjunction with IEC 61918:2018.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 61918:2018, *Industrial communication networks – Installation of communication networks in industrial premises*

The normative references of IEC 61918:2018, Clause 2, apply.

NOTE For profile specific normative references, see Clauses A.2 and B.2.

3 Terms, definitions, symbols and abbreviations

For the purposes of this document, the terms and definitions given in IEC 61918 apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

NOTE For profile specific terms, definitions and abbreviated terms see Clauses A.3 and B.3.

4 CPF 20: Overview of installation profiles

CPF 20 consists of two Communication Profiles as specified in IEC 61784-2.

The installation requirements for CP 20/1 (ADS-net/μΣNETWORK-1000[1]) are specified in Annex A.

The installation requirements for CP 20/2 (ADS-net/NX[1]) are specified in Annex B.

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1 ADS-net, ADS-net/μΣNETWORK-1000 and ADS-net/NX are used to describe this document.

5 Installation profile conventions

The numbering of the clauses and subclauses in the annexes of this document corresponds to the numbering of IEC 61918 main clauses and subclauses.

The annex clauses and subclauses of this document supplement, modify, or replace the respective clauses and subclauses in IEC 61918.

Where there is no corresponding subclause of IEC 61918 in the normative annexes in this document, the subclause of IEC 61918 applies without modification.

The annex heading letter represents the installation profile assigned in Clause 4.

The annex heading number shall represent the corresponding numbering of IEC 61918.

EXAMPLE "Subclause A.4.4" in IEC 61784-5-20 means that CP 20/1 specifies the Subclause 4.4 of IEC 61918.

All main clauses of IEC 61918 are cited and apply in full unless otherwise stated in each normative installation profile annex.

If all subclauses of a (sub)clause are omitted, then the corresponding IEC 61918 (sub)clause applies.

If in a (sub)clause it is written "Not applicable", then the corresponding IEC 61918 (sub)clause does not apply.

If in a (sub)clause it is written "*Addition:*", then the corresponding IEC 61918 (sub)clause applies with the additions written in the profile.

If in a (sub)clause it is written "*Replacement:*", then the text provided in the profile replaces the text of the corresponding IEC 61918 (sub)clause.

NOTE A replacement can also comprise additions.

If in a (sub)clause it is written "*Modification:*", then the corresponding IEC 61918 (sub)clause applies with the modifications written in the profile.

If all (sub)clauses of a (sub)clause are omitted but in this (sub)clause it is written "*(Sub)clause x has addition:"* (or "*replacement*:") or "(Sub)clause x is not applicable.", then (sub)clause x becomes valid as declared and all the other corresponding IEC 61918 (sub)clauses apply.

6 Conformance to installation profiles

Each installation profile within this document includes part of IEC 61918:2018. It may also include defined additional specifications.

A statement of compliance to an installation profile of this document shall be stated[2] as either

Compliance to IEC 61784-5-20:—[3] for CP 20/n <name> or

2 In accordance with ISO/IEC Directives.

3 The date should not be used when the edition number is used.

Compliance to IEC 61784-5-20 (Ed.1.0) for CP 20/n <name>

where the name within the angle brackets < > is optional and the angle brackets are not to be included. The n within CP 20/n shall be replaced by the profile number 1 to 2.

NOTE The name may be the name of the profile, for example ADS-net/μΣNETWORK-1000 or ADS-net/NX.

If the name is a trade name then the permission of the trade name holder shall be required. Product standards shall not include any conformity assessment aspects (including quality management provisions), neither normative nor informative, other than provisions for product testing (evaluation and examination).

Annex A
(normative)

CP 20/1 (ADS-net/μΣNETWORK-1000) specific installation profile

A.1 Installation profile scope

Addition:

This annex specifies the installation profile for Communication Profile CP 20/1 (ADS-net/μΣNETWORK-1000). The CP 20/1 is specified in IEC 61784-2.

A.2 Normative references

Addition:

IEC 60603-7-2:2010, *Connectors for electronic equipment – Part 7-2: Detail specification for 8-way, unshielded, free and fixed connectors, for data transmissions with frequencies up to 100 MHz*

IEC 60603-7-3:2010, *Connectors for electronic equipment – Part 7-3: Detail specification for 8-way, shielded, free and fixed connectors, for data transmission with frequencies up to 100 MHz*

IEC 60603-7-4:2010, *Connectors for electronic equipment – Part 7-4: Detail specification for 8-way, unshielded, free and fixed connectors, for data transmissions with frequencies up to 250 MHz*

IEC 60603-7-5:2010, *Connectors for electronic equipment – Part 7-5: Detail specification for 8-way, shielded, free and fixed connectors, for data transmissions with frequencies up to 250 MHz*

IEC 60793-2-50:2015, *Optical fibres – Part 2-50: Product specifications – Sectional specification for class B single-mode fibres*

A.3 Installation profile terms, definitions, and abbreviated terms

A.3.1 Terms and definitions

A.3.2 Abbreviated terms

A.3.3 Conventions for installation profiles

Not applicable.

A.4 Installation planning

A.4.1 General

A.4.1.1 Objective

A.4.1.2 Cabling in industrial premises

A.4.1.3 The planning process

A.4.1.4 Special requirements for CPs

Not applicable.

A.4.1.5 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.2 Planning requirements

A.4.2.1 Safety

A.4.2.1.1 General

A.4.2.1.2 Electric safety

A.4.2.1.3 Functional safety

Not applicable.

A.4.2.1.4 Intrinsic safety

Not applicable.

A.4.2.1.5 Safety of optical fibre communication systems

A.4.2.2 Security

A.4.2.3 Environmental considerations and EMC

A.4.2.3.1 Description methodology

A.4.2.3.2 Use of the described environment to produce a bill of material

A.4.2.4 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.3 Network capabilities

A.4.3.1 Network topology

A.4.3.1.1 Common description

A.4.3.1.2 Basic physical topologies for passive networks

Not applicable.

A.4.3.1.3 Basic physical topologies for active networks

Addition:

The ring topology shall be used for CP 20/1 active networks.

A.4.3.1.4 Combination of basic topologies

A.4.3.1.5 Specific requirements for CPs

Not applicable.

A.4.3.1.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.3.2 Network characteristics

A.4.3.2.1 General

A.4.3.2.2 Network characteristics for balanced cabling not based on Ethernet

Not applicable.

A.4.3.2.3 Network characteristics for balanced cabling based on Ethernet

Replacement:

Table A.1 provides values based on the template given in IEC 61918:2018, Table 2.

Table A.1 – Network characteristics for balanced cabling based on Ethernet

Characteristic	CP 20/1
Supported data rates (Mbit/s)	10, 100, and 1 000
Supported channel length (m) [b]	100
Number of connections in the channel (max.) [a] [b]	6
Patch cord length (m) [a]	100
Channel class per ISO/IEC 11801-3 (min.) [b]	D
Cable category per ISO/IEC 11801-3 (min.) [c]	5
Connecting HW category per ISO/IEC 11801-3 (min.)	5
Cable types	As needed for application

[a] See 4.4.3.2.

[b] For the purpose of this table, the channel definitions of ISO/IEC 11801-3 are applicable.

[c] For additional information, see IEC 61156 series.

A.4.3.2.4 Network characteristics for optical fibre cabling

Replacement:

Table A.2 provides values based on the template given in IEC 61918:2018, Table 3.

Table A.2 – Network characteristics for optical fibre cabling

CP 20/1		
Optical fibre type	**Description**	
Single mode silica	Bandwidth (MHz) or equivalent at λ (nm)	OS1 1310 and 1550
	Minimum length (m)	0
	Maximum length[a] (m)	40 000
	Maximum channel insertion loss/optical power budget (dB)	According to ISO/IEC 11801
	Connecting hardware	See A.4.4.2.5
Multimode silica	Bandwidth (MHz x km) or equivalent at λ (nm)	OM2 850
	Minimum length (m)	0
	Maximum length[a] (m)	550
	Maximum channel insertion loss/optical power budget (dB)	According to ISO/IEC 11801
	Connecting hardware	See A.4.4.2.5
[a] This value is reduced by connections, splices and bends in accordance with Formula (1) in IEC 61918:2018, 4.4.3.4.1.		

A.4.3.2.5 Specific network characteristics

Not applicable.

A.4.3.2.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4 Selection and use of cabling components

A.4.4.1 Cable selection

A.4.4.1.1 Common description

A.4.4.1.2 Copper cables

Replacement:

Table A.3 provides values based on the template given in IEC 61918:2018, Table 4.

Table A.3 – Information relevant to copper cable: fixed cables

Characteristic	CP 20/1
Nominal impedance of cable (tolerance)	100 Ω ± 15 Ω
DCR of conductors	< 9,38 Ω/100 m
DCR of shield	Not defined
Number of conductors	8
Shielding	Shielded/Unshielded
Colour code for conductor	WH/OG, OG, WH/GN, BU, WH/BU, GN, WH/BN, BN
Jacket colour requirements	No requirement
Cable types	No requirement
Jacket material	No requirement
Resistance to harsh environment (e.g. UV, oil resist, LS0H)	No requirement
Agency ratings	No requirement

Replacement:

Table A.4 provides values based on the template given in IEC 61918:2018, Table 5.

Table A.4 – Information relevant to copper cable: cords

Characteristic	CP 20/1
Nominal impedance of cable (tolerance)	100 Ω ± 15 Ω
DCR of conductors	< 9,38 Ω/100 m
DCR of shield	Not defined
Number of conductors	8
Length	≤ 100 m
Shielding	Shielded/Unshielded
Colour code for conductor	WH/OG, OG, WH/GN, BU, WH/BU, GN, WH/BN, BN
Jacket colour requirements	No requirement
Jacket material	No requirement
Resistance to harsh environment (e.g. UV, oil resist, LS0H)	No requirement
Agency ratings	No requirement

A.4.4.1.3 Cables for wireless installation

Not applicable.

A.4.4.1.4 Optical fibre cables

Replacement:

Table A.5 provides values based on the template given in IEC 61918:2018, Table 6.

Table A.5 – Information relevant to optical fibre cables

Characteristic	9..10/125 µm single mode silica	50/125 µm multi mode silica	62,5/125 µm multi mode silica	900/1 000 µm step index POF	200/230 µm step index hard clad silica
Standard	IEC 60793-2-50; Type B1	IEC 60793-2-10; Type A1a	-	-	-
Attenuation per km (650 nm)	-	-	-	-	-
Attenuation per km (820 nm)	-	-	-	-	-
Attenuation per km (1310 nm)	≤ 0,5 dB	≤ 1,5 dB	-	-	-
Number of optical fibres	1 or 2	2	-	-	-
Jacket colour requirements	[a]	[a]	-	-	-
Jacket material	[a]	[a]	-	-	-
Resistance to harsh environment (e.g. UV, oil resist, LS0H)	[a]	[a]	-	-	-
Breakout (Y/N)	Yes	Yes	-	-	-

[a] Application dependant.

A.4.4.1.5 Special purpose balanced and optical fibre cables

A.4.4.1.6 Specific requirements for CPs

Not applicable.

A.4.4.1.7 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4.2 Connecting hardware selection

A.4.4.2.1 Common description

A.4.4.2.2 Connecting hardware for balanced cabling CPs based on Ethernet

Replacement:

Table A.6 provides values based on the template given in IEC 61918:2018, Table 7.

Table A.6 – Connectors for balanced cabling CPs based on Ethernet

	IEC 60603-7-x[a]		IEC 61076-3-106[b]		IEC 61076-3-117[b]	IEC 61076-2-101
	(shielded)	(unshielded)	Variant 1	Variant 6	Variant 14	M12-4 with D-coding
CP 20/1	Yes	Yes	No	No	No	No

[a] For IEC 60603-7-x, the connector selection is based on the desired channel performance.

[b] Housings to protect connectors.

A.4.4.2.3 Connecting hardware for copper cabling CPs not based on Ethernet

Not applicable.

A.4.4.2.4 Connecting hardware for wireless installation

Not applicable.

A.4.4.2.5 Connecting hardware for optical fibre cabling

Replacement:

Table A.7 provides values based on the template given in IEC 61918:2018, Table 9.

Table A.7 – Optical fibre connecting hardware

	IEC 61754-2	IEC 61754-4	IEC 61754-24	IEC 61754-20	IEC 61754-22	Others
	BFOC/2,5	**SC**	**SC-RJ**	**LC**	**F-SMA**	
CP 20/1	No	No	No	Yes	No	No
NOTE The IEC 61754 series defines the optical fibre connector mechanical interfaces; performance specifications for optical fibre connectors terminated to specific fibre types are standardized in the IEC 61753 series.						

Replacement:

Table A.8 provides values based on the template given in IEC 61918:2018, Table 10.

Table A.8 – Relationship between FOC and fibre types (CP 20/1)

	Fibre type					
	9..10/125 µm single mode silica	**50/125 µm multi mode silica**	**62,5/125 µm multi mode silica**	**900/1 000 µm step index POF**	**200/230 µm step index hard clad silica**	**Others**
BFOC/2,5	No	No	No	No	No	No
SC	No	No	No	No	No	No
SC-RJ	No	No	No	No	No	No
LC	Yes	Yes	No	No	No	No
F-SMA	No	No	No	No	No	No
Others	No	No	No	No	No	No

A.4.4.2.6 Specific requirements for CPs

Not applicable.

A.4.4.2.7 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4.3 Connections within a channel/permanent link

A.4.4.3.1 Common description

A.4.4.3.2 Balanced cabling connections and splices for CPs based on Ethernet

A.4.4.3.2.1 Common description

A.4.4.3.2.2 Connections minimum distance

A.4.4.3.2.3 Balanced cabling splices

A.4.4.3.2.4 Balanced cabling bulkhead connections

A.4.4.3.2.5 Balanced cabling J-J coupler (JJ adaptor)

A.4.4.3.3 Copper cabling connections and splices for CPs not based on Ethernet

Not applicable.

A.4.4.3.4 Optical fibre cabling connections and splices for CPs based on Ethernet

A.4.4.3.4.1 Common description

A.4.4.3.4.2 Optical fibre splices

A.4.4.3.4.3 Optical fibre bulkhead connections

A.4.4.3.4.4 Optical fibre J-J couplers (or adaptors)

A.4.4.3.5 Optical fibre cabling connections and splices for CPs not based on Ethernet

Not applicable.

A.4.4.3.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4.4 Terminators

Not applicable.

A.4.4.5 Device location and connection

A.4.4.5.1 Common description

A.4.4.5.2 Specific requirements for CPs

Not applicable.

A.4.4.5.3 Specific requirements for wireless installation

Not applicable.

A.4.4.5.4 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4.6 Coding and labelling

A.4.4.6.1 Common description

A.4.4.6.2 Additional requirements for CPs

Not applicable.

A.4.4.6.3 Specific requirements for CPs

Not applicable.

A.4.4.6.4 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4.7 Earthing and bonding of equipment and devices and shielded cabling

A.4.4.7.1 Common description

A.4.4.7.1.1 Basic requirements

A.4.4.7.1.2 Planner tasks

A.4.4.7.1.3 Methods for controlling potential differences in the earth system

A.4.4.7.1.4 Selection of the earthing and bonding systems

A.4.4.7.2 Bonding and earthing of enclosures and pathways

A.4.4.7.2.1 Equalisation and earthing conductor sizing and length

A.4.4.7.2.2 Bonding straps and sizing

A.4.4.7.2.3 Surface preparation and methods

A.4.4.7.2.4 Bonding and earthing

A.4.4.7.3 Earthing methods

A.4.4.7.3.1 Equipotential

A.4.4.7.3.2 Star

A.4.4.7.3.3 Earthing of equipment (devices)

A.4.4.7.3.4 Copper bus bars

A.4.4.7.4 Shield earthing

A.4.4.7.4.1 Non-earthing or parallel RC

A.4.4.7.4.2 Direct

A.4.4.7.4.3 Derivatives of direct and parallel RC

A.4.4.7.5 Specific requirements for CPs

Not applicable.

A.4.4.7.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4.8 Storage and transportation of cables

A.4.4.8.1 Common description

A.4.4.8.2 Specific requirements for CPs

Not applicable.

A.4.4.8.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4.9 Routing of cables

A.4.4.9.1 Common description

A.4.4.9.2 Cable routing of assemblies

A.4.4.9.3 Requirements for cable routing inside enclosures

A.4.4.9.4 Cable routing inside buildings

A.4.4.9.5 Cable routing outside and between buildings

A.4.4.9.6 Installing redundant communication cables

A.4.4.10 Separation of circuits

A.4.4.11 Mechanical protection of cabling components

A.4.4.11.1 Common description

A.4.4.11.2 Specific requirements for CPs

Not applicable.

A.4.4.11.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.4.12 Installation in special areas

A.4.4.12.1 Common description

A.4.4.12.2 Specific requirements for CPs

Not applicable.

A.4.4.12.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.4.5 Cabling planning documentation

A.4.5.1 Common description

A.4.5.2 Cabling planning documentation for CPs

A.4.5.3 Network certification documentation

A.4.5.4 Cabling planning documentation for generic cabling in accordance with ISO/IEC 11801-3

A.4.6 Verification of cabling planning specification

A.5 Installation implementation

A.5.1 General requirements

A.5.1.1 Common description

A.5.1.2 Installation of CPs

A.5.1.3 Installation of generic cabling in industrial premises

A.5.2 Cable installation

A.5.2.1 General requirements for all cabling types

A.5.2.1.1 Storage and installation

A.5.2.1.2 Protecting communication cables against potential mechanical damage

Replacement:

Table A.9 provides values based on the template given in IEC 61918:2018, Table 18.

Table A.9 – Parameters for balanced cables

Characteristic		Value
Mechanical force	Minimum bending radius, single bending (mm)	[a]
	Bending radius, multiple bending (mm)	[a]
	Pull forces (N)	[a]
	Permanent tensile forces (N)	[a]
	Maximum lateral forces (N/cm)	[a]
	Temperature range during installation (°C)	[a]
[a] Depending on cable type; see manufacture’s data sheet.		

Replacement:

Table A.10 provides values based on the template given in IEC 61918:2018, Table 19.

Table A.10 – Parameters for silica optical fibre cables

Characteristic		Value
Mechanical force	Minimum bending radius, single bending (mm)	50 (during installation) 30 (after installation)
	bending radius, multiple bending (mm)	Vendor specified
	Pull forces (N)	[a]
	Permanent tensile forces (N)	[a]
	Maximum lateral forces (N/cm)	[a]
	Temperature range during installation (°C)	Vendor specified

[a] Depending on cable type; see manufacturer's data sheet.

A.5.2.1.3 Avoid forming loops

A.5.2.1.4 Torsion (twisting)

A.5.2.1.5 Tensile strength (on installed cables)

A.5.2.1.6 Bending radius

A.5.2.1.7 Pull force

A.5.2.1.8 Fitting strain relief

A.5.2.1.9 Installing cables in cabinet and enclosures

A.5.2.1.10 Installation on moving parts

A.5.2.1.11 Cable crush

A.5.2.1.12 Installation of continuous flexing cables

A.5.2.1.13 Additional instructions for the installation of optical fibre cables

A.5.2.1.13.1 Use cable pulling tools

A.5.2.1.13.2 Cautions for handling optical fibre cables

A.5.2.1.13.3 Keeping plugs clean

A.5.2.1.13.4 Attenuation change under load

A.5.2.1.13.5 Strain relief

A.5.2.1.13.6 EMC ruggedness

A.5.2.1.13.7 Crush resistance

A.5.2.2 Installation and routing

A.5.2.2.1 Common description

A.5.2.2.2 Separation of circuits

A.5.2.3 Specific requirements for CPs

Not applicable.

A.5.2.4 Specific requirements for wireless installation

Not applicable.

A.5.2.5 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.5.3 Connector installation

A.5.3.1 Common description

A.5.3.2 Shielded connectors

A.5.3.3 Unshielded connectors

A.5.3.4 Specific requirements for CPs

Not applicable.

A.5.3.5 Specific requirements for wireless installation

Not applicable.

A.5.3.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.5.4 Terminator installation

Not applicable.

A.5.5 Device installation

A.5.5.1 Common description

A.5.5.2 Specific requirements for CPs

Not applicable.

A.5.6 Coding and labelling

A.5.6.1 Common description

A.5.6.2 Specific requirements for CPs

Not applicable.

A.5.7 Earthing and bonding of equipment and devices and shield cabling

A.5.7.1 Common description

A.5.7.2 Bonding and earthing of enclosures and pathways

A.5.7.2.1 Equalisation and earthing conductor sizing and length

A.5.7.2.2 Bonding straps and sizing

A.5.7.2.3 Surface preparation and methods

A.5.7.3 Earthing methods

A.5.7.3.1 Equipotential

A.5.7.3.2 Star

A.5.7.3.3 Earthing of equipment (devices)

A.5.7.3.3.1 Non-earthing or parallel RC

A.5.7.3.3.2 Direct

A.5.7.3.3.3 Installing copper bus bars

A.5.7.4 Shield earthing methods

A.5.7.4.1 General

A.5.7.4.2 Parallel RC

A.5.7.4.3 Direct

A.5.7.4.4 Derivatives of direct and parallel RC

A.5.7.5 Specific requirements for CPs

Not applicable.

A.5.7.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.5.8 As-implemented cabling documentation

A.6 Installation verification and installation acceptance test

A.6.1 General

A.6.2 Installation verification

A.6.2.1 General

A.6.2.2 Verification according to cabling planning documentation

A.6.2.3 Verification of earthing and bonding

A.6.2.3.1 General

A.6.2.3.2 Specific requirements for earthing and bonding

Not applicable.

A.6.2.4 Verification of shield earthing

A.6.2.5 Verification of cabling system

A.6.2.5.1 Verification of cable routing

A.6.2.5.2 Verification of cable protection and proper strain relief

A.6.2.6 Cable selection verification

A.6.2.6.1 Common description

A.6.2.6.2 Specific requirements for CPs

Not applicable.

A.6.2.6.3 Specific requirements for wireless installation

Not applicable.

A.6.2.7 Connector verification

A.6.2.7.1 Common description

A.6.2.7.2 Specific requirements for CPs

Not applicable.

A.6.2.7.3 Specific requirements for wireless installation

Not applicable.

A.6.2.8 Connection verification

A.6.2.8.1 Common description

A.6.2.8.2 Number of connections and connectors

A.6.2.8.3 Wire mapping

A.6.2.9 Terminator verification

Not applicable.

A.6.2.10 Coding and labelling verification

A.6.2.10.1 Common description

A.6.2.10.2 Specific coding and labelling verification requirements

Not applicable.

A.6.2.11 Verification report

A.6.3 Installation acceptance test

A.6.3.1 General

A.6.3.2 Acceptance test of Ethernet based cabling

A.6.3.2.1 Validation of balanced cabling for CPs based on Ethernet

A.6.3.2.1.1 Common description

A.6.3.2.1.2 Transmission performance test parameters

A.6.3.2.1.3 Specific requirements for CPs based on Ethernet

Not applicable.

A.6.3.2.2 Validation of optical fibre cabling for CPs based on Ethernet

A.6.3.2.2.1 Common description

A.6.3.2.2.2 Specific requirements for optical fibre cabling CPs

Not applicable.

A.6.3.2.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

A.6.3.3 Acceptance test of non-Ethernet based cabling

Not applicable.

A.6.3.4 Specific requirements for wireless installation

Not applicable.

A.6.3.5 Acceptance test report

A.7 Installation administration

Subclause 7.8 is not applicable.

A.8 Installation maintenance and installation troubleshooting

Subclause 8.4 is not applicable.

Annex B
(normative)

CP 20/2 (ADS-net/NX) specific installation profile

B.1 Installation profile scope

Addition:

This annex specifies the installation profile for Communication Profile CP 20/2 (ADS-net/NX). The CP 20/2 is specified in IEC 61784-2.

B.2 Normative references

Addition:

IEC 60603-7-2:2010, *Connectors for electronic equipment – Part 7-2: Detail specification for 8-way, unshielded, free and fixed connectors, for data transmissions with frequencies up to 100 MHz*

IEC 60603-7-3:2010, *Connectors for electronic equipment – Part 7-3: Detail specification for 8-way, shielded, free and fixed connectors, for data transmission with frequencies up to 100 MHz*

IEC 60603-7-4:2010, *Connectors for electronic equipment – Part 7-4: Detail specification for 8-way, unshielded, free and fixed connectors, for data transmissions with frequencies up to 250 MHz*

IEC 60603-7-5:2010, *Connectors for electronic equipment – Part 7-5: Detail specification for 8-way, shielded, free and fixed connectors, for data transmissions with frequencies up to 250 MHz*

IEC 60793-2-50:2015, *Optical fibres – Part 2-50: Product specifications – Sectional specification for class B single-mode fibres*

B.3 Installation profile terms, definitions, and abbreviated terms

B.3.1 Terms and definitions

B.3.2 Abbreviated terms

B.3.3 Conventions for installation profiles

Not applicable.

B.4 Installation planning

B.4.1 General

B.4.1.1 Objective

B.4.1.2 Cabling in industrial premises

B.4.1.3 The planning process

B.4.1.4 Special requirements for CPs

Not applicable.

B.4.1.5 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.2 Planning requirements

B.4.2.1 Safety

B.4.2.1.1 General

B.4.2.1.2 Electric safety

B.4.2.1.3 Functional safety

Not applicable.

B.4.2.1.4 Intrinsic safety

Not applicable.

B.4.2.1.5 Safety of optical fibre communication systems

B.4.2.2 Security

B.4.2.3 Environmental considerations and EMC

B.4.2.3.1 Description methodology

B.4.2.3.2 Use of the described environment to produce a bill of material

B.4.2.4 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.3 Network capabilities

B.4.3.1 Network topology

B.4.3.1.1 Common description

B.4.3.1.2 Basic physical topologies for passive networks

Not applicable.

B.4.3.1.3 Basic physical topologies for active networks

Addition:

Additional topologies increase network availability. CP 20/2 supports redundant active linear bus topologies.

B.4.3.1.4 Combination of basic topologies

Not applicable.

B.4.3.1.5 Specific requirements for CPs

Not applicable.

B.4.3.1.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.3.2 Network characteristics

B.4.3.2.1 General

B.4.3.2.2 Network characteristics for balanced cabling not based on Ethernet

Not applicable.

B.4.3.2.3 Network characteristics for balanced cabling based on Ethernet

Replacement:

Table B.1 provides values based on the template given in IEC 61918:2018, Table 2.

Table B.1 – Network characteristics for balanced cabling based on Ethernet

Characteristic	CP 20/2
Supported data rates (Mbit/s)	10, 100, and 1 000
Supported channel length (m) [b]	100
Number of connections in the channel (max.) [a] [b]	6
Patch cord length (m) [a]	100
Channel class per ISO/IEC 11801-3 (min.) [b]	D
Cable category per ISO/IEC 11801-3 (min.) [c]	5
Connecting HW category per ISO/IEC 11801-3 (min.)	5
Cable types	As needed for application

[a] See 4.4.3.2.

[b] For the purpose of this table, the channel definitions of ISO/IEC 11801-3 are applicable.

[c] For additional information, see IEC 61156 series.

B.4.3.2.4 Network characteristics for optical fibre cabling

Replacement:

Table B.2 provides values based on the template given in IEC 61918:2018, Table 3.

Table B.2 – Network characteristics for optical fibre cabling

CP 20/2		
Optical fibre type	**Description**	
Single mode silica	Bandwidth (MHz) or equivalent at λ (nm)	OS1 1 310 and 1 550
	Minimum length (m)	0
	Maximum length[a] (m)	40 000
	Maximum channel insertion loss/optical power budget (dB)	According to ISO/IEC 11801
	Connecting hardware	See A.4.4.2.5
Multimode silica	Bandwidth (MHz x km) or equivalent at λ (nm)	OM2 850
	Minimum length (m)	0
	Maximum length[a] (m)	550
	Maximum channel insertion loss/optical power budget (dB)	According to ISO/IEC 11801
	Connecting hardware	See A.4.4.2.5
[a] This value is reduced by connections, splices and bends in accordance with Formula (1) in IEC 61918:2018, 4.4.3.4.1.		

B.4.3.2.5 Specific network characteristics

Not applicable.

B.4.3.2.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.4 Selection and use of cabling components

B.4.4.1 Cable selection

B.4.4.1.1 Common description

B.4.4.1.2 Copper cables

Replacement:

Table B.3 provides values based on the template given in IEC 61918:2018, Table 4.

Table B.3 – Information relevant to copper cable: fixed cables

Characteristic	CP 20/2
Nominal impedance of cable (tolerance)	100 Ω ± 15 Ω
DCR of conductors	< 9,38 Ω/100 m
DCR of shield	Not defined
Number of conductors	8
Shielding	Shielded/Unshielded
Colour code for conductor	WH/OG, OG, WH/GN, BU, WH/BU, GN, WH/BN, BN
Jacket colour requirements	No requirement
Cable types	No requirement
Jacket material	No requirement
Resistance to harsh environment (e.g. UV, oil resist, LS0H)	No requirement
Agency ratings	No requirement

Replacement:

Table B.4 provides values based on the template given in IEC 61918:2018, Table 5.

Table B.4 – Information relevant to copper cable: cords

Characteristic	CP 20/2
Nominal impedance of cable (tolerance)	100 Ω ± 15 Ω
DCR of conductors	< 9,38 Ω/100 m
DCR of shield	Not defined
Number of conductors	8
Length	≤ 100 m
Shielding	Shielded/Unshielded
Colour code for conductor	WH/OG, OG, WH/GN, BU, WH/BU, GN, WH/BN, BN
Jacket colour requirements	No requirement
Jacket material	No requirement
Resistance to harsh environment (e.g. UV, oil resist, LS0H)	No requirement
Agency ratings	No requirement

B.4.4.1.3 Cables for wireless installation

Not applicable.

B.4.4.1.4 Optical fibre cables

Replacement:

Table B.5 provides values based on the template given in IEC 61918:2018, Table 6.

Table B.5 – Information relevant to optical fibre cables

Characteristic	9..10/125 µm single mode silica	50/125 µm multi mode silica	62,5/125 µm multi mode silica	900/1 000 µm step index POF	200/230 µm step index hard clad silica
Standard	IEC 60793-2-50; Type B1	IEC 60793-2-10; Type A1a	-	-	-
Attenuation per km (650 nm)	-	-	-	-	-
Attenuation per km (820 nm)	-	-	-	-	-
Attenuation per km (1310 nm)	≤ 0,5 dB	≤ 1,5 dB	-	-	-
Number of optical fibres	1 or 2	2	-	-	-
Jacket colour requirements	-	[a]	-	-	-
Jacket material	-	[a]	-	-	-
Resistance to harsh environment (e.g. UV, oil resist, LS0H)	-	[a]	-	-	-
Breakout (Y/N)	Yes	Yes	-	-	-
[a] Application dependant.					

B.4.4.1.5 Special purpose balanced and optical fibre cables

B.4.4.1.6 Specific requirements for CPs

Not applicable.

B.4.4.1.7 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.4.2 Connecting hardware selection

B.4.4.2.1 Common description

B.4.4.2.2 Connecting hardware for balanced cabling CPs based on Ethernet

Replacement:

Table B.6 provides values based on the template given in IEC 61918:2018, Table 7.

Table B.6 – Connectors for balanced cabling CPs based on Ethernet

	IEC 60603-7-x[a]		IEC 61076-3-106[b]		IEC 61076-3-117[b]	IEC 61076-2-101
	(shielded)	(unshielded)	Variant 1	Variant 6	Variant 14	M12-4 with D-coding
CP 20/2	Yes	Yes	No	No	No	No
[a] For IEC 60603-7-x, the connector selection is based on the desired channel performance.						
[b] Housings to protect connectors.						

B.4.4.2.3 Connecting hardware for copper cabling CPs not based on Ethernet

Not applicable.

B.4.4.2.4 Connecting hardware for wireless installation

Not applicable.

B.4.4.2.5 Connecting hardware for optical fibre cabling

Replacement:

Table provides values based on the template given in IEC 61918:2018, Table 9.

Table B.7 – Optical fibre connecting hardware

	IEC 61754-2	IEC 61754-4	IEC 61754-24	IEC 61754-20	IEC 61754-22	Others
	BFOC/2,5	**SC**	**SC-RJ**	**LC**	**F-SMA**	
CP 20/2	No	No	No	Yes	No	No
NOTE The IEC 61754 series defines the optical fibre connector mechanical interfaces; performance specifications for optical fibre connectors terminated to specific fibre types are standardized in the IEC 61753 series.						

Replacement:

Table B.8 provides values based on the template given in IEC 61918:2018, Table 10.

Table B.8 – Relationship between FOC and fibre types (CP 20/2)

	Fibre type					
	9..10/125 µm single mode silica	**50/125 µm multi mode silica**	**62.5/125 µm multi mode silica**	**900/1 000 µm step index POF**	**200/230 µm step index hard clad silica**	**Others**
BFOC/2,5	No	No	No	No	No	No
SC	No	No	No	No	No	No
SC-RJ	No	No	No	No	No	No
LC	Yes	Yes	No	No	No	No
F-SMA	No	No	No	No	No	No
Others	No	No	No	No	No	No

B.4.4.2.6 Specific requirements for CPs

Not applicable.

B.4.4.2.7 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.4.3 Connections within a channel/permanent link

B.4.4.3.1 Common description

B.4.4.3.2 Balanced cabling connections and splices for CPs based on Ethernet

B.4.4.3.2.1 Common description

B.4.4.3.2.2 Connections minimum distance

B.4.4.3.2.3 Balanced cabling splices

B.4.4.3.2.4 Balanced cabling bulkhead connections

B.4.4.3.2.5 Balanced cabling J-J coupler (JJ adaptor)

B.4.4.3.3 Copper cabling connections and splices for CPs not based on Ethernet

Not applicable.

B.4.4.3.4 Optical fibre cabling connections and splices for CPs based on Ethernet

B.4.4.3.4.1 Common description

B.4.4.3.4.2 Optical fibre splices

B.4.4.3.4.3 Optical fibre bulkhead connections

B.4.4.3.4.4 Optical fibre J-J couplers (or adaptors)

B.4.4.3.5 Optical fibre cabling connections and splices for CPs not based on Ethernet

Not applicable.

B.4.4.3.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.4.4 Terminators

Not applicable.

B.4.4.5 Device location and connection

B.4.4.5.1 Common description

B.4.4.5.2 Specific requirements for CPs

Not applicable.

B.4.4.5.3 Specific requirements for wireless installation

Not applicable.

B.4.4.5.4 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.4.6 Coding and labelling

B.4.4.6.1 Common description

B.4.4.6.2 Additional requirements for CPs

Not applicable.

B.4.4.6.3 Specific requirements for CPs

Not applicable.

B.4.4.6.4 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.4.7 Earthing and bonding of equipment and devices and shielded cabling

B.4.4.7.1 Common description

B.4.4.7.1.1 Basic requirements

B.4.4.7.1.2 Planner tasks

B.4.4.7.1.3 Methods for controlling potential differences in the earth system

B.4.4.7.1.4 Selection of the earthing and bonding systems

B.4.4.7.2 Bonding and earthing of enclosures and pathways

B.4.4.7.2.1 Equalisation and earthing conductor sizing and length

B.4.4.7.2.2 Bonding straps and sizing

B.4.4.7.2.3 Surface preparation and methods

B.4.4.7.2.4 Bonding and earthing

B.4.4.7.3 Earthing methods

B.4.4.7.3.1 Equipotential

B.4.4.7.3.2 Star

B.4.4.7.3.3 Earthing of equipment (devices)

B.4.4.7.3.4 Copper bus bars

B.4.4.7.4 Shield earthing

B.4.4.7.4.1 Non-earthing or parallel RC

B.4.4.7.4.2 Direct

B.4.4.7.4.3 Derivatives of direct and parallel RC

B.4.4.7.5 Specific requirements for CPs

Not applicable.

B.4.4.7.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.4.8 Storage and transportation of cables

B.4.4.8.1 Common description

B.4.4.8.2 Specific requirements for CPs

Not applicable.

B.4.4.8.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.4.9 Routing of cables

B.4.4.9.1 Common description

B.4.4.9.2 Cable routing of assemblies

B.4.4.9.3 Requirements for cable routing inside enclosures

B.4.4.9.4 Cable routing inside buildings

B.4.4.9.5 Cable routing outside and between buildings

B.4.4.9.6 Installing redundant communication cables

B.4.4.10 Separation of circuits

B.4.4.11 Mechanical protection of cabling components

B.4.4.11.1 Common description

B.4.4.11.2 Specific requirements for CPs

Not applicable.

B.4.4.11.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.4.12 Installation in special areas

B.4.4.12.1 Common description

B.4.4.12.2 Specific requirements for CPs

Not applicable.

B.4.4.12.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.4.5 Cabling planning documentation

B.4.5.1 Common description

B.4.5.2 Cabling planning documentation for CPs

B.4.5.3 Network certification documentation

B.4.5.4 Cabling planning documentation for generic cabling in accordance with ISO/IEC 11801-3

B.4.6 Verification of cabling planning specification

B.5 Installation implementation

B.5.1 General requirements

B.5.1.1 Common description

B.5.1.2 Installation of CPs

B.5.1.3 Installation of generic cabling in industrial premises

B.5.2 Cable installation

B.5.2.1 General requirements for all cabling types

B.5.2.1.1 Storage and installation

B.5.2.1.2 Protecting communication cables against potential mechanical damage

Replacement:

Table B.9 provides values based on the template given in IEC 61918:2018, Table 18.

Table B.9 – Parameters for balanced cables

Characteristic		Value
Mechanical force	Minimum bending radius, single bending (mm)	[a]
	Bending radius, multiple bending (mm)	[a]
	Pull forces (N)	[a]
	Permanent tensile forces (N)	[a]
	Maximum lateral forces (N/cm)	[a]
	Temperature range during installation (°C)	[a]

[a] Depending on cable type: see manufacture's data sheet.

Replacement:

Table B.10 provides values based on the template given in IEC 61918:2018, Table 19.

Table B.10 – Parameters for silica optical fibre cables

Characteristic		Value
Mechanical force	Minimum bending radius, single bending (mm)	50 (during installation) 30 (after installation)
	bending radius, multiple bending (mm)	Vendor specified
	Pull forces (N)	[a]
	Permanent tensile forces (N)	[a]
	Maximum lateral forces (N/cm)	[a]
	Temperature range during installation (°C)	Vendor specified
[a] Depending on cable type; see manufacturer's data sheet.		

B.5.2.1.3 Avoid forming loops

B.5.2.1.4 Torsion (twisting)

B.5.2.1.5 Tensile strength (on installed cables)

B.5.2.1.6 Bending radius

B.5.2.1.7 Pull force

B.5.2.1.8 Fitting strain relief

B.5.2.1.9 Installing cables in cabinet and enclosures

B.5.2.1.10 Installation on moving parts

B.5.2.1.11 Cable crush

B.5.2.1.12 Installation of continuous flexing cables

B.5.2.1.13 Additional instructions for the installation of optical fibre cables

B.5.2.1.13.1 Use cable pulling tools

B.5.2.1.13.2 Cautions for handling optical fibre cables

B.5.2.1.13.3 Keeping plugs clean

B.5.2.1.13.4 Attenuation change under load

B.5.2.1.13.5 Strain relief

B.5.2.1.13.6 EMC ruggedness

B.5.2.1.13.7 Crush resistance

B.5.2.2 Installation and routing

B.5.2.2.1 Common description

B.5.2.2.2 Separation of circuits

B.5.2.3 Specific requirements for CPs

Not applicable.

B.5.2.4 Specific requirements for wireless installation

Not applicable.

B.5.2.5 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.5.3 Connector installation

B.5.3.1 Common description

B.5.3.2 Shielded connectors

B.5.3.3 Unshielded connectors

B.5.3.4 Specific requirements for CPs

Not applicable.

B.5.3.5 Specific requirements for wireless installation

Not applicable.

B.5.3.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.5.4 Terminator installation

Not applicable.

B.5.5 Device installation

B.5.5.1 Common description

B.5.5.2 Specific requirements for CPs

Not applicable.

B.5.6 Coding and labelling

B.5.6.1 Common description

B.5.6.2 Specific requirements for CPs

Not applicable.

B.5.7 Earthing and bonding of equipment and devices and shield cabling

B.5.7.1 Common description

B.5.7.2 Bonding and earthing of enclosures and pathways

B.5.7.2.1 Equalisation and earthing conductor sizing and length

B.5.7.2.2 Bonding straps and sizing

B.5.7.2.3 Surface preparation and methods

B.5.7.3 Earthing methods

B.5.7.3.1 Equipotential

B.5.7.3.2 Star

B.5.7.3.3 Earthing of equipment (devices)

B.5.7.3.3.1 Non-earthing or parallel RC

B.5.7.3.3.2 Direct

B.5.7.3.3.3 Installing copper bus bars

B.5.7.4 Shield earthing methods

B.5.7.4.1 General

B.5.7.4.2 Parallel RC

B.5.7.4.3 Direct

B.5.7.4.4 Derivatives of direct and parallel RC

B.5.7.5 Specific requirements for CPs

Not applicable.

B.5.7.6 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.5.8 As-implemented cabling documentation

B.6 Installation verification and installation acceptance test

B.6.1 General

B.6.2 Installation verification

B.6.2.1 General

B.6.2.2 Verification according to cabling planning documentation

B.6.2.3 Verification of earthing and bonding

B.6.2.3.1 General

B.6.2.3.2 Specific requirements for earthing and bonding

Not applicable.

B.6.2.4 Verification of shield earthing

B.6.2.5 Verification of cabling system

B.6.2.5.1 Verification of cable routing

B.6.2.5.2 Verification of cable protection and proper strain relief

B.6.2.6 Cable selection verification

B.6.2.6.1 Common description

B.6.2.6.2 Specific requirements for CPs

Not applicable.

B.6.2.6.3 Specific requirements for wireless installation

Not applicable.

B.6.2.7 Connector verification

B.6.2.7.1 Common description

B.6.2.7.2 Specific requirements for CPs

Not applicable.

B.6.2.7.3 Specific requirements for wireless installation

Not applicable.

B.6.2.8 Connection verification

B.6.2.8.1 Common description

B.6.2.8.2 Number of connections and connectors

B.6.2.8.3 Wire mapping

B.6.2.9 Terminator verification

Not applicable.

B.6.2.10 Coding and labelling verification

B.6.2.10.1 Common description

B.6.2.10.2 Specific coding and labelling verification requirements

B.6.2.11 Verification report

B.6.3 Installation acceptance test

B.6.3.1 General

B.6.3.2 Acceptance test of Ethernet based cabling

B.6.3.2.1 Validation of balanced cabling for CPs based on Ethernet

B.6.3.2.1.1 Common description

B.6.3.2.1.2 Transmission performance test parameters

B.6.3.2.1.3 Specific requirements for CPs based on Ethernet

Not applicable.

B.6.3.2.2 Validation of optical fibre cabling for CPs based on Ethernet

B.6.3.2.2.1 Common description

B.6.3.2.2.2 Specific requirements for optical fibre cabling CPs

Not applicable.

B.6.3.2.3 Specific requirements for generic cabling in accordance with ISO/IEC 11801-3

B.6.3.3 Acceptance test of non-Ethernet based cabling

Not applicable.

B.6.3.4 Specific requirements for wireless installation

Not applicable.

B.6.3.5 Acceptance test report

B.7 Installation administration

Subclause 7.8 is not applicable.

B.8 Installation maintenance and installation troubleshooting

Subclause 8.4 is not applicable.

Bibliography

Addition:

[40] MSTC/JOP 1101:1999, *Specifications for Autonomous Decentralized Protocol R3.0*

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